

FreeCAD: Sketcher Constraints

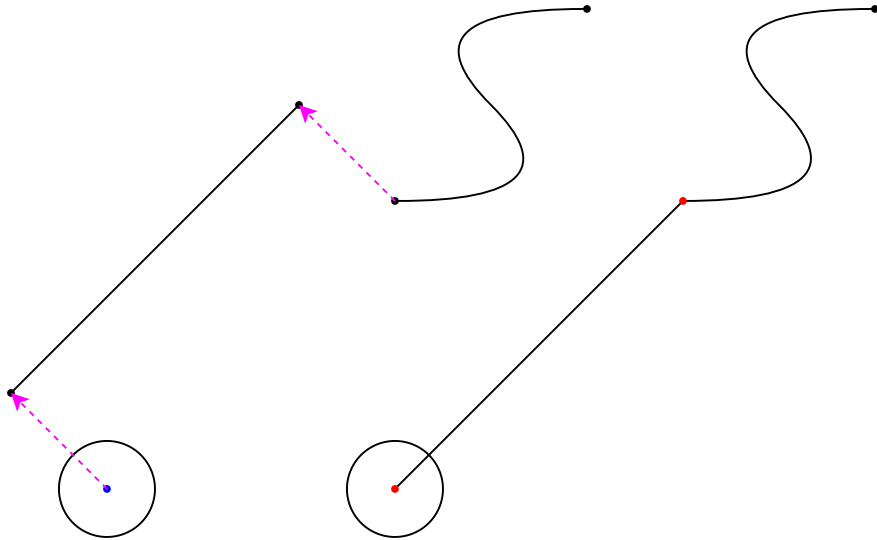


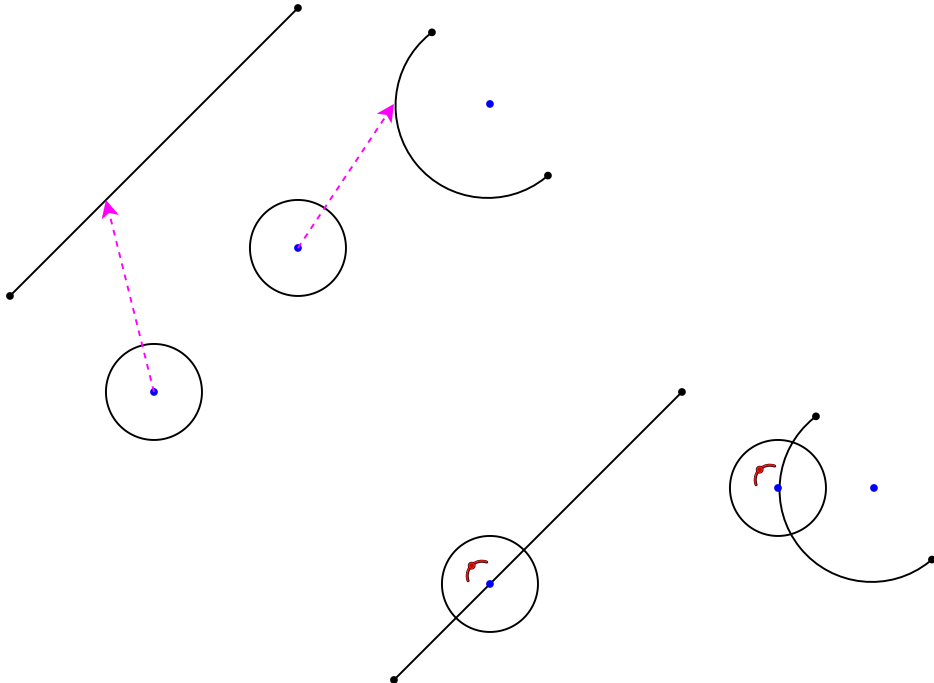
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|------------------------|----------------------|
| 1) Coincident | 6) Tangent/Collinear |
| 2) Horizontal/Vertical | 7) Equal |
| 3) Point on Object | 8) Symmetric |
| 4) Parallel | 9) Block |
| 5) Perpendicular | |

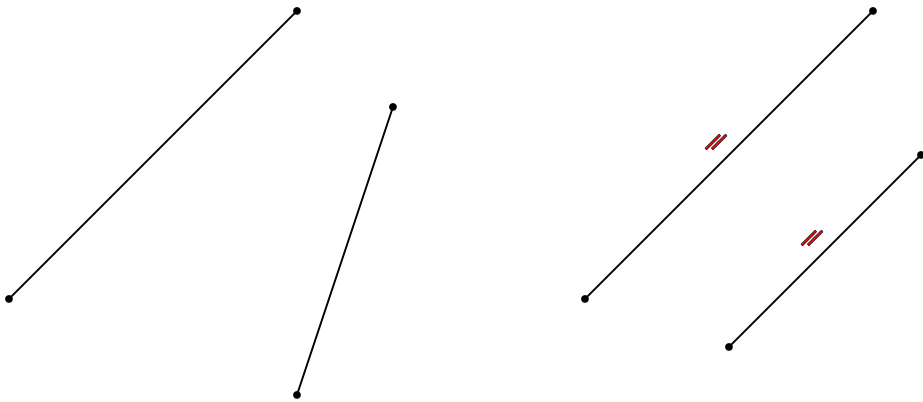
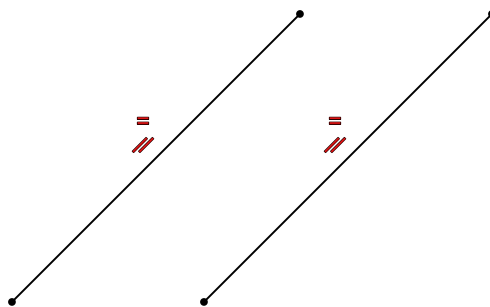
What is a constraint?

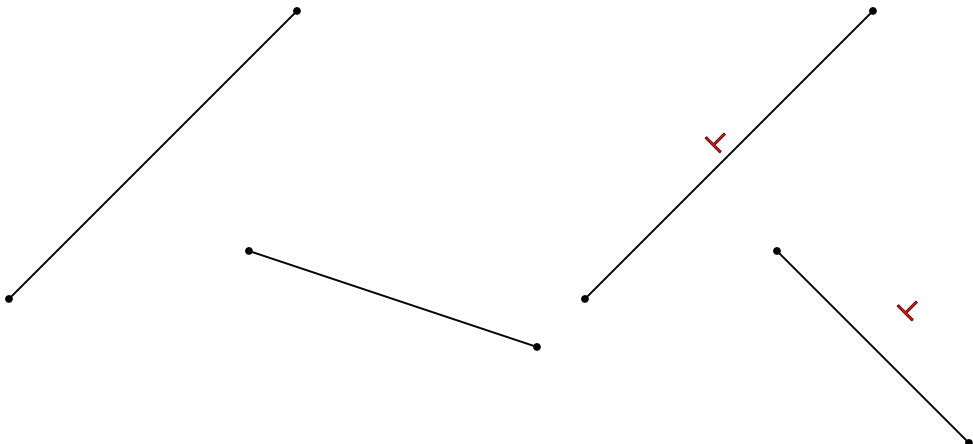
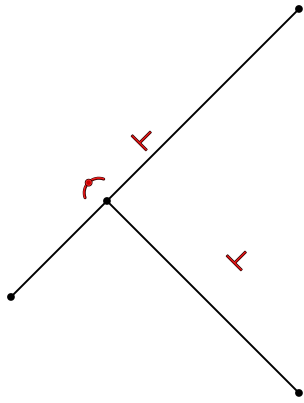
Constraints ensure points or lines with a geometric relationship stay consistent throughout the sketch.

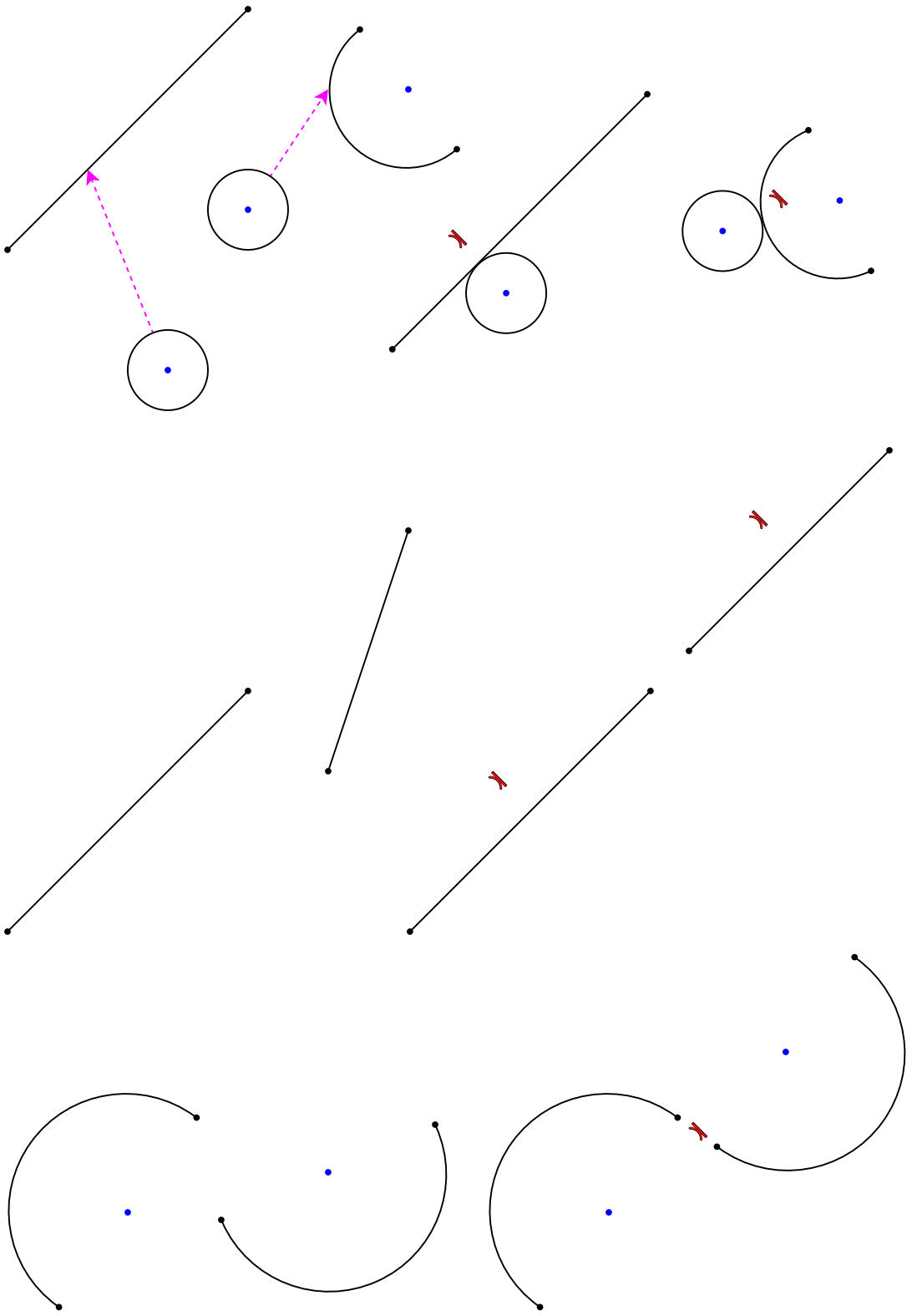


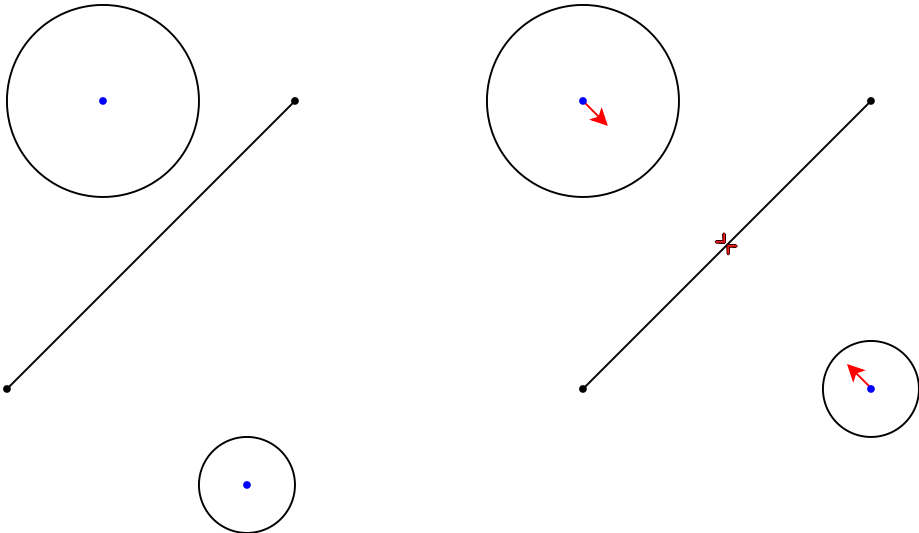
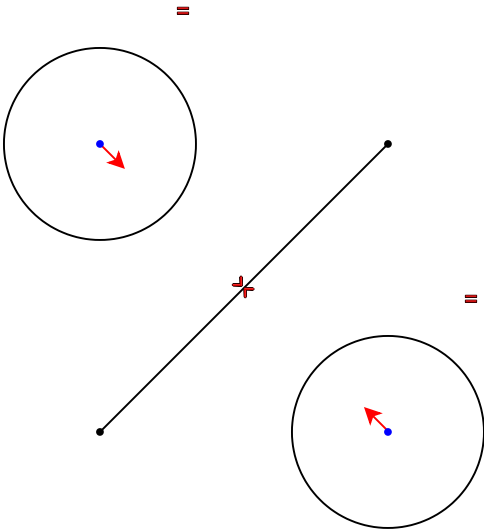
Constraint	Description & Example
 Coincident (C)	Attach a point to another point, line, or arc. If both points are unconstrained the second point selected will move to the first point selected. 

Constraint	Description & Example
 Horizontal/Vertical (A)	Ensures the selected line always stays horizontal or vertical on this sketch plane, determined by its closest alignment.
 Point on Object (None)	A point attached to another point, line, or arc. This constraint is automatically created by Coincident following FreeCAD version 1.0. 

Constraint	Description & Example
 Parallel (P)	Orients two lines to be parallel.  Use in conjunction with Equal to make both lines the same length. 

Constraint	Description & Example
 Perpendicular (N)	Orients two lines to be perpendicular.  Use in conjunction with Coincident to create a T-shape. 

Constraint	Description & Example
 Tangent/Collinear (T)	Attach an arc or ellipse to a line or another arc/ellipse. Can also be used to make two arcs/ellipses tangent and two lines collinear. 

Constraint	Description & Example
 Equal (E)	Ensures two lines share the same length or an arc or circle share the same radius. If both circles or arcs are unconstrained the second shape selected will match the first shape selected.
 Symmetric (S)	Orients two points to share the same distance and angle from a line. Preselect two points and a line before invoking this command.  Use Equal to ensure figures symmetric across the line are mirrored. Use Distance to set the distance from the line of symmetry. 

Constraint	Description & Example
 Block (K, B)	Prevents lines from changing slope/angle, length or location.